

Quiz 1

Calcular norma L_2 , $\|x\|_2$: $x = [-6, -6, -7]$

$$L_2 : \sqrt{\sum_{i=1}^n x_i^2} \longrightarrow \sqrt{(-6)^2 + (-6)^2 + (-7)^2} = \sqrt{121} = 11$$

Calcular norma espectral $\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)}$: $A = \begin{bmatrix} 7 & -4 \\ 0 & 0 \end{bmatrix}$
 $A^T = \begin{bmatrix} 7 & 0 \\ -4 & 0 \end{bmatrix}$

$$A^T A = \begin{bmatrix} 7 & 0 \\ -4 & 0 \end{bmatrix} \begin{bmatrix} 7 & -4 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{cases} a = 7 \cdot 7 + 0 \cdot (-4) = 49 \\ b = 7 \cdot (-4) = -28 \\ c = (-4) \cdot 7 = -28 \\ d = (-4) \cdot (-4) = 16 \end{cases}$$

$$A^T A = \begin{bmatrix} 49 & -28 \\ -28 & 16 \end{bmatrix}$$

$$V.A : \det(A - I\lambda) = 0$$

$$\det \begin{bmatrix} 49 - \lambda & -28 \\ -28 & 16 - \lambda \end{bmatrix} = 0$$

$$(49 - \lambda)(16 - \lambda) - (-28 \cdot -28) = 0$$

$$\left. \begin{array}{l} 49 \cdot 16 = 784 \\ 49 \cdot (-\lambda) = -49\lambda \\ 16 \cdot (-\lambda) = -16\lambda \\ \lambda^2 \end{array} \right\} \begin{array}{l} -784 \\ \downarrow \\ \lambda^2 - 49\lambda - 16\lambda + 784 - 784 = 0 \\ \downarrow \\ \lambda^2 - 65\lambda = 0 \end{array}$$

$$\lambda^2 - 65\lambda = 0$$

$$\lambda(\lambda - 65) = 0 \quad \begin{cases} \lambda_1 = 0 \\ \lambda_2 = 65 \end{cases}$$

$$\lambda_1 = 65$$

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import numpy as np

X = np.array([[9, -6, -6], [-2, -4, -7], [-7, 5, 5]])

v = np.array([-2, -6, -7])

$$X = \begin{bmatrix} 9 & -6 & -6 \\ -2 & -4 & -7 \\ -7 & 5 & 5 \end{bmatrix}$$

Y = X + v[:, np.newaxis]

prints out
for elements

print(Y[1, 2])

array 1 dimension

v = [-2 -6 -7]

$$v[:, \text{np.newaxis}] \rightarrow \begin{bmatrix} -2 \\ -6 \\ -7 \end{bmatrix}$$

Broadcasting

$$\begin{aligned} \text{de } \mathbb{R}^m &\rightarrow \mathbb{R}^{m \times m} \\ \mathbb{R}^3 &\rightarrow \mathbb{R}^{3 \times 1} \end{aligned}$$

$$\begin{bmatrix} 9 & -6 & -6 \\ -2 & -4 & -7 \\ -7 & 5 & 5 \end{bmatrix} + \begin{bmatrix} -2 \\ -6 \\ -7 \end{bmatrix} = \begin{bmatrix} 7 & -8 & -8 \\ -8 & -10 & -13 \\ -14 & -2 & -2 \end{bmatrix}$$

X v Y

$$\text{print}(Y[1, 2]) = \boxed{-13}$$

python den indexen 0

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import numpy as np

v = np.array([1, 2, 3, 4, 5, 6, 7, ..., 20])

x = v[v > 4] → [5, 6, 7, ..., 20]

z = x[1:6] → [6, 7, 8, 9, 10]

print(z[-3]) : En python los índices negativos se usan para contar desde atrás

z[-1] = 10

z[2] = 9

z[-3] = 8